Thermodynamic description of the Al–Mg system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Al–Mg binary system as calculated with pycalphad from the SGTE dataset AlMg-12Tan.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Al) + β, lies at 723.9 K.

1. Introduction

The Al–Mg system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as intermetallic rich, peritectic. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of Y. Tang, Y. Du, L. Zhang, X. Yuan, G. Kaptay, Thermochim. Acta, 527, 131-42(2012). The calculations use pycalphad 0.11.2 at 1 atm.

The melting point of Al reproduced by the dataset is 933.5 K. The melting point of Mg reproduced by the dataset is 923.0 K.

2. Phase diagram

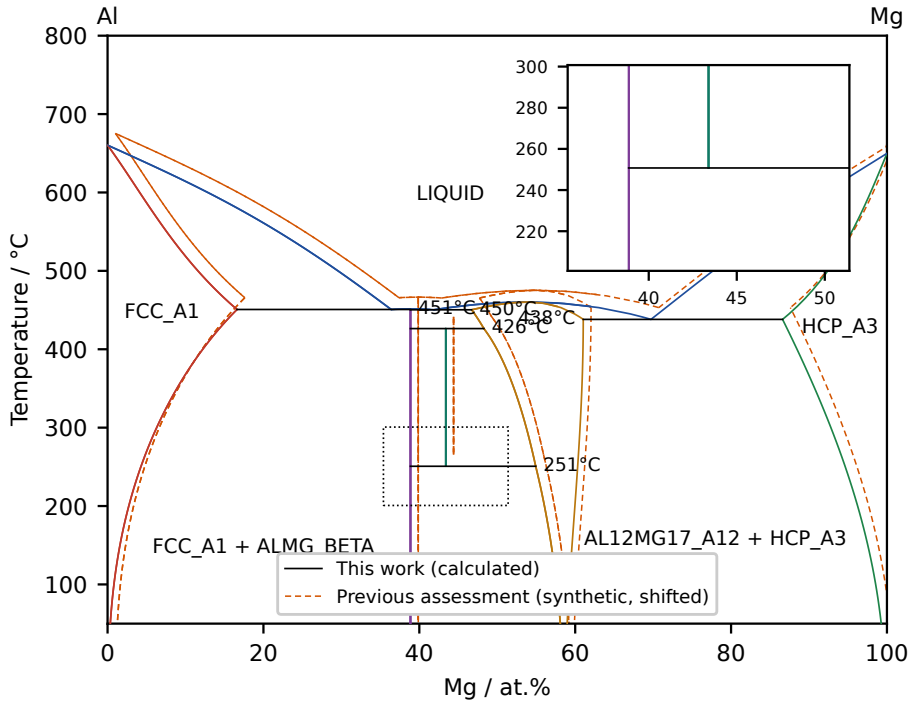


Fig. 1. Calculated Al–Mg phase diagram and a dashed, deliberately shifted second assessment (inset: enlarged view near the lowest invariant).

The dataset places a eutectic (L → (Al) + β) at 723.9 K; the composition of L is 36.4 at.% Mg. We obtain 723.5 K for the eutectic reaction (L → β + Al12Mg17), where L has 41.9 at.%. The dataset places a eutectic (L → Al12Mg17 + (Mg)) at 711.2 K; the composition of L is 69.7 at.%. At 699.5 K the peritectoid β + Al12Mg17 → ε takes place; the ε phase then contains 43.4 at.%. The eutectoid reaction ε → β + Al12Mg17 is calculated at 523.9

K, with ε at 43.4 at.% Mg. Al12Mg17 melts congruently at 733.4 K and 54.2 at.% Mg. β melts congruently at 724.7 K and 38.9 at.% Mg.

Table 1. Invariant reactions in the Al–Mg system (calculated).

Reaction	Type	T (°C)	at.% Mg, 1	at.% Mg, 2	at.% Mg, 3
L → (Al) + β	eutectic	450.7	36.4	16.6	38.9
L → β + Al12Mg17	eutectic	450.4	41.9	38.9	46.7
L → Al12Mg17 + (Mg)	eutectic	438.1	69.7	61.0	86.6
β + Al12Mg17 → ε	peritectoid	426.4	38.9	48.3	43.4
ε → β + Al12Mg17	eutectoid	250.7	43.4	38.9	54.9

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Thermodynamic properties

At 1000 K the integral mixing enthalpy of the liquid has its minimum, −2.34 kJ/mol, at 52.0 at.% Mg. The calculated enthalpy of formation of β at 298.15 K is −4.53 kJ/mol per mole of atoms.

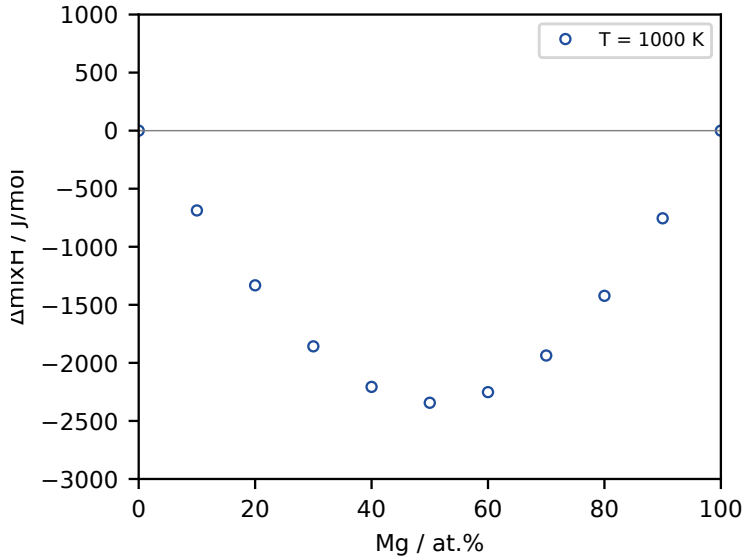


Fig. 2. Calculated mixing enthalpy of the liquid at 1000 K.

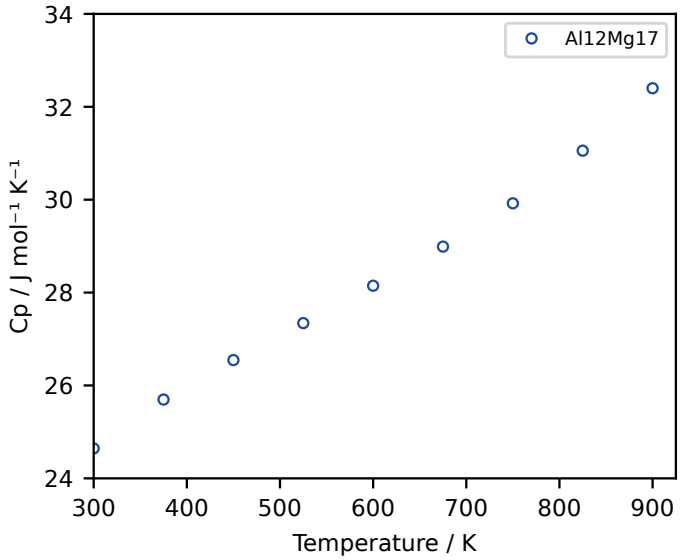


Fig. 3. Calculated heat capacity of Al12Mg17.

Table 2. Calculated formation properties of the intermediate phases (Al–Mg).

Phase	at.% Mg	ΔfH(298 K) / kJ mol⁻¹	S(298 K) / J mol⁻¹ K⁻¹
Al12Mg17	56.3	−3.52	31.24
β	38.9	−4.53	28.76
ε	43.4	−3.99	29.94

Reference: stable elements (SER) at 298.15 K.

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.